Shayan | CSE 325 | Microcontroller
Class - 0 | 1

CT Syllabus

Chapter-II : real mode; protected mode; segment offset math; selector descriptor math (From Barry's book)

Chapter-0 : Numbering system, RAM vs ROM, Inside a computer, bus

Chapter-1 : Microprocessor vs Microcontroller, criteria for choosing a microcontroller, diagram of 8051 with brief description

Chapter-2 : PSW, Stack, Short terms like : EQU, END, ORG

Chapter-3 : LOOP, DJNZ, JZ, JK, Time Calculation, Short jump address calculation

CT Date : 17 তারিখ 2026

Barry's Book
Chapter-III
Addressing Mode

MOV Dest, Src

↗ 8 bit Addressing Mode

1. Register Addressing Mode : $\boxed{\text{MOV AX, BX}}$ (Register to Register Movement)
2. Immediate : $\boxed{\text{MOV CH, 3AH}}$ (সরাসরি Data বসানো)
3. Direct : $\boxed{\text{MOV [1234], AX}}$ (Segment Address দিয়ে Data বসানো হয়]

2 2য় half

Segment Address Calculation

Physical Address : $\boxed{(\text{Segment} \times 10) + \text{Offset}}$

MOV [1234], AX

offset

"AX এর value [1234]'র location-এ বসবে"

④ Register Indirect: $\boxed{\text{MOV [BX], CL}}$ (Register এর মধ্যে Memory location)

BX is a register

[ ] denotes memory location

[BX] denotes location using the register

⑤ Base-Plus Index: $\boxed{\text{MOV [BX+SI], BP}}$

BX = Base Register

SI = Source Index

⑥ Register Relative : $\boxed{\text{MOV CL, [BX+4]}}$

⑦ Base Relative Plus Index:

MOV ARRAY [BX, SI], DX	(with array)
MOV, [BX+SI+4], CL	(without array)

⑧ Scaled Index: $\boxed{\text{MOV [EBX+ 2×ESI], AX}}$

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Book : 128 | Barry's Book

(22) DS = D200H, BX = 0300H, DI = 400H

Determine mem address (assume real mode operation)

(a) MOV AL, [1234 H] (Direct)

(b) MOV EAX, [BX]

(c) MOV [DI], AL (Direct)

(a) MOV AL, [1234H]

Memory Address = DS × 10 + 1234

= (0200H × 10) + 1234

= 2000 + 1234

= ~~3324~~ 3234

(b) MOV EAX, [BX]

Memory Address = DS × 10 + 300

= 2300

(c) MOV [DI], AL

DS × 10H + ~~BX~~ DI

= (0200H × 10H) + ~~B~~ 400H

= 2400H

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DS = 1100H
BX = 0200H
LIST = 0250H
SI = 0500H

(a) MOV LIST[SI], EDX

= DS x 10H + LIST + BX + SI

= (1100H x 10H) + 0250H + 0200H + 0500H

= 11950H

(b) MOV CL, LIST [BX+SI]

= DS x 10H + BX + SI + LIST

= (1100H x 10H) + 0200H + 0500H + 0250H

= 11000 + 200 + 500

= ~~11700H~~ 11950

(c) MOV CH, [BX+SI]

= DS x 10H + BX + SI

= (1100H x 10H) + 0200H + 0500H

= 11700H

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DS vs SS which one to use? ← Marut's book

↗ Addressing Mode Description

Figure 3.3 (page 102) / 32bit

Register Array

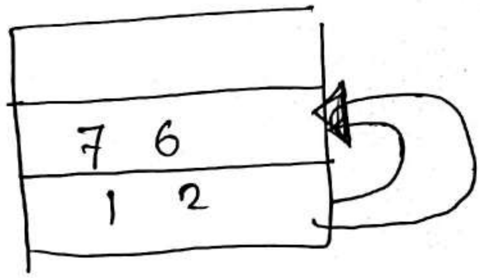

Figure 3.4 (page 103)

EAX	3333	34	56
EBX			

Figure 3.5 (Direct) page 106